EEG-VAE: Full Paper Title

Anonymous Authors[1]

Abstract

Ton résumé ici (150–200 mots).

1. Introduction

Electroencephalography (EEG) measures the brain's electrical activity through electrodes placed on the scalp, capturing oscillatory dynamics across a broad frequency spectrum with millisecond temporal resolution (Teplan, 2002; Müller-Putz, 2020). These properties make EEG a central modality in brain-computer interfaces (BCIs), with applications spanning motor imagery (Pfurtscheller & Neuper, 2001), imagined speech decoding (Elwasify et al., 2026), cognitive monitoring, and clinical diagnosis. Yet EEG signals are inherently noisy, non-stationary, and highly subject-specific, which makes decoding a notoriously difficult problem. Traditional approaches relied on hand-crafted features and shallow classifiers, but deep learning has substantially improved performance across tasks (Roy et al., 2019). Despite this progress, most methods remain task-specific and require large labeled datasets for each new application—a bottleneck that severely limits their clinical and real-world deployment.

The success of large pretrained models in natural language processing and computer vision has inspired a growing body of work on *EEG foundation models*—models trained on diverse, unlabeled EEG corpora and then adapted to downstream tasks via lightweight probing (Alain & Bengio, 2017). Several such models have been proposed recently, leveraging the Transformer architecture (Vaswani et al., 2017) with masked reconstruction as the pretraining objective: EEGPT (Wang et al., 2024), CBraMod (Wang et al., 2025a), REVE (Ouahidi et al., 2026), and CSBrain (Zhou et al., 2026). In parallel, state space models (SSMs) such as Mamba (Gu & Dao, 2024) and Mamba-2 (Dao & Gu, 2024) have emerged as efficient alternatives to attention-based architectures, with linear complexity in sequence length—a desirable property for long EEG recordings. EEGMamba (Wang et al., 2025b), FEMBA (Tegon et al., 2025), LUNA (Döner et al., 2025), and LuMamba (Broustail et al., 2026) explore this direction, but their pretraining objectives remain anchored to pixel-space reconstruction in the input signal domain.

We argue that reconstruction in the raw signal space is a suboptimal pretraining objective for learning *semantic* EEG representations. First, raw EEG contains substantial low-level noise and physiological artifacts that do not carry task-relevant information; a model trained to reconstruct them may allocate significant capacity to irrelevant details. Second, patch-based masked reconstruction encourages local pattern matching rather than higher-level semantic structure. VAE-based approaches for EEG (Zhao et al., 2024; You et al., 2025) alleviate some of these issues by learning a compressed latent space, but they have not been scaled to large corpora or combined with structured self-supervised pretraining. In other domains—audio compression with EnCodec (Défossez et al., 2022), image tokenization with RQ-VAE (Lee et al., 2022), and medical imaging with MedVAE (Varma et al., 2025)—residual vector quantization has demonstrated that discrete, hierarchical codebooks provide compact yet expressive representations that benefit downstream modeling.

In this work, we introduce **EEG-RQVAE**, a foundation model for EEG based on a Residual Quantized Variational Autoencoder trained with a two-stage pretraining pipeline designed to learn increasingly semantic representations. In the first stage, the model is trained to reconstruct a clean, augmented view of the EEG from a corrupted input, leveraging denoising as an implicit regularizer while a patch discriminator enforces signal fidelity. In the second stage, we adapt the encoder using a Joint-Embedding Predictive Architecture (Assran et al., 2023) in latent space, predicting the representations

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

of masked spatial and temporal regions from visible context—encouraging the model to capture the structured, predictive dependencies that underlie brain dynamics. Throughout both stages, we use bidirectional Mamba blocks as the backbone, enabling efficient modeling of long EEG sequences while respecting the temporal symmetry of brain signals.

Our main contributions are:

- **EEG-RQVAE**: a residual vector-quantized VAE for multi-channel EEG that produces discrete, hierarchical token sequences suitable for both generative and discriminative downstream use.

- **A two-stage pretraining pipeline**: combining denoising reconstruction (Stage 1) with latent JEPA-style adaptation (Stage 2) to progressively enforce semantic representation learning.

- **An analysis of SSM architectures for EEG**: we investigate the relevance of bidirectional Mamba blocks for spatio-temporal EEG modeling and compare with Transformer baselines.

- **A fair evaluation benchmark**: we evaluate learned representations with a standardized linear probing protocol across multiple BCI tasks, with a particular focus on imagined speech decoding, and compare against recent EEG foundation models under identical conditions.

2. Method

2.1. Background

- introduction to ssm and mamba / mamba 2, the relevance of BiMamba blocks. - VQ-VAE and RVQ - Jepa

2.2. Training pipeline overview

- 2 stages : reconstruction and jepa (explain why)

2.3. Architecture

- Details over the architectural choices

2.4. Pretraining : Stage 1

- explain the goal of this stage and the different steps / losses involved. - developp on the different augmentations and corruptions used and their relevance for EEG data. (the denoising and regularization effects)

2.5. Pretraining : Stage 2

- explain the goal of this stage and the different steps / losses involved.

3. Experiments

3.1. Datasets and preprocessing

- table which lists the different datasets with BCI task, name, number of channels, montage, duration of events, number of classes, number of subjects and number of samples.

3.2. Pretraining

3.3. Evaluation on downstream tasks

- metrics and downstream protocol used for evaluation. - we used the code of the models from original papers and integrated it in our pipeline with a linear probe with the same structure but adapted for each model. - table with the results across the different models avg + std (best in bold and second best undelined)

4. Limitations and future work

- I would like to focus on a more applied modality which is imagined speech decoding. Many applications in BCI. Comparison between phonemes and words decoding. ...

5. Conclusion

References

Alain, G. and Bengio, Y. Understanding intermediate layers using linear classifier probes, 2017. URL https://openreview.net/forum?id=ryF7rTqgl.

Assran, M., Duval, Q., Misra, I., Bojanowski, P., Vincent, P., Rabbat, M., LeCun, Y., and Ballas, N. Self-supervised learning from images with a joint-embedding predictive architecture, 2023. URL https://arxiv.org/abs/2301.08243.

Balestriero, R. and LeCun, Y. Lejepa: Provable and scalable self-supervised learning without the heuristics, 11 2025.

Broustail, D., Tegon, A., Ingolfsson, T., Li, Y., and Benini, L. Lumamba: Latent unified mamba for electrode topology-invariant and efficient eeg modeling, 03 2026.

Coelli, S., Calcagno, A., Cassani, C. M., Temporiti, F., Reali, P., Gatti, R., Galli, M., and Bianchi, A. M. Selecting methods for a modular eeg pre-processing pipeline: An objective comparison. *Biomedical Signal Processing and Control*, 90:105830, 2024. ISSN 1746-8094. doi: https://doi.org/10.1016/j.bspc.2023.105830. URL https://www.sciencedirect.com/science/article/pii/S1746809423012636.

Dai, S., Xiao, H., Yu, S., and Ye, Q. Autoregressive visual decoding from eeg signals, 2026. URL https://arxiv.org/abs/2602.22555.

Dao, T. and Gu, A. Transformers are SSMs: Generalized models and efficient algorithms through structured state space duality. In Salakhutdinov, R., Kolter, Z., Heller, K., Weller, A., Oliver, N., Scarlett, J., and Berkenkamp, F. (eds.), *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pp. 10041–10071. PMLR, 21–27 Jul 2024. URL https://proceedings.mlr.press/v235/dao24a.html.

Döner, B., Ingolfsson, T. M., Benini, L., and Li, Y. LUNA: Efficient and topology-agnostic foundation model for EEG signal analysis. In *1st ICML Workshop on Foundation Models for Structured Data*, 2025. URL https://openreview.net/forum?id=TWaw5qtKQf.

Défossez, A., Copet, J., Synnaeve, G., and Adi, Y. High fidelity neural audio compression, 2022. URL https://arxiv.org/abs/2210.13438.

Elwasify, F., Shaaban, E., and Abdelmoneem, R. Eeg imagined speech neuro-signal preprocessing and deep learning classification. *Scientific Reports*, 16, 03 2026. doi: 10.1038/s41598-026-39395-6.

Goodfellow, I. J., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., and Bengio, Y. Generative adversarial nets. In Ghahramani, Z., Welling, M., Cortes, C., Lawrence, N., and Weinberger, K. (eds.), *Advances in Neural Information Processing Systems*, volume 27. Curran Associates, Inc., 2014. URL https://proceedings.neurips.cc/paper_files/paper/2014/file/f033ed80deb0234979a61f95710dbe25-Paper.pdf.

Gu, A. and Dao, T. Mamba: Linear-time sequence modeling with selective state spaces. In *First Conference on Language Modeling*, 2024. URL https://openreview.net/forum?id=tEYskw1VY2.

Gu, A., Goel, K., and Ré, C. Efficiently modeling long sequences with structured state spaces, 10 2021.

Lee, D., Kim, C., Kim, S., Cho, M., and Han, W.-S. Autoregressive Image Generation using Residual Quantization . In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 11513–11522, Los Alamitos, CA, USA, June 2022. IEEE Computer Society. doi: 10.1109/CVPR52688.2022.01123. URL https://doi.ieeecomputersociety.org/10.1109/CVPR52688.2022.01123.

Ma, H., Chen, Y., Zhao, W., Yang, J., Ji, Y., Xu, X., Liu, X., Jing, H., Liu, S., and Yang, G. A mamba foundation model for time series forecasting. *CoRR*, abs/2411.02941, 2024. URL https://doi.org/10.48550/arXiv.2411.02941.

Müller-Putz, G. R. Chapter 18 - electroencephalography. In Ramsey, N. F. and del R. Millán, J. (eds.), *Brain-Computer Interfaces*, volume 168 of *Handbook of Clinical Neurology*, pp. 249–262. Elsevier, 2020. doi: https://doi.org/10.1016/B978-0-444-63934-9.00018-4. URL https://www.sciencedirect.com/science/article/pii/B9780444639349000184.

Ouahidi, Y. E., Lys, J., Thölke, P., Farrugia, N., Pasdeloup, B., Gripon, V., Jerbi, K., and Lioi, G. REVE: A foundation model for EEG - adapting to any setup with large-scale pretraining on 25,000 subjects. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2026. URL https://openreview.net/forum?id=ZeFMtRBy4Z.

Pfurtscheller, G. and Neuper, C. Motor imagery and direct brain-computer communication. *Proceedings of the IEEE*, 89(7): 1123–1134, 2001. doi: 10.1109/5.939829.

Roy, Y., Banville, H., Carneiro de Albuquerque, I. M., Gramfort, A., Falk, T., and Faubert, J. Deep learning-based electroencephalography analysis: a systematic review. *Journal of Neural Engineering*, 16, 08 2019. doi: 10.1088/1741-2552/ab260c.

Samiee, S. and Baillet, S. Time-resolved phase-amplitude coupling in neural oscillations. *NeuroImage*, 159:270–279, 2017. ISSN 1053-8119. doi: https://doi.org/10.1016/j.neuroimage.2017.07.051. URL https://www.sciencedirect.com/science/article/pii/S1053811917306195.

Tegon, A., Ingolfsson, T. M., Wang, X., Benini, L., and Li, Y. Femba: Efficient and scalable eeg analysis with a bidirectional mamba foundation model. *CoRR*, abs/2502.06438, February 2025. URL https://doi.org/10.48550/arXiv.2502.06438.

Teplan, M. Fundamental of eeg measurement. *MEASUREMENT SCIENCE REVIEW*, 2, 01 2002.

Varma, M., Kumar, A., van der Sluijs, R., Ostmeier, S., Blankemeier, L., Chambon, P., Bluethgen, C., Prince, J., Langlotz, C., and Chaudhari, A. Medvae: Efficient automated interpretation of medical images with large-scale generalizable autoencoders, 2025. URL https://arxiv.org/abs/2502.14753.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L. u., and Polosukhin, I. Attention is all you need. In Guyon, I., Luxburg, U. V., Bengio, S., Wallach, H., Fergus, R., Vishwanathan, S., and Garnett, R. (eds.), *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc., 2017. URL https://proceedings.neurips.cc/paper_files/paper/2017/file/3f5ee243547dee91fbd053c1c4a845aa-Paper.pdf.

Wang, G., Liu, W., He, Y., Xu, C., Ma, L., and Li, H. Eegpt: Pretrained transformer for universal and reliable representation of eeg signals. In Globerson, A., Mackey, L., Belgrave, D., Fan, A., Paquet, U., Tomczak, J., and Zhang, C. (eds.), *Advances in Neural Information Processing Systems*, volume 37, pp. 39249–39280. Curran Associates, Inc., 2024. doi: 10.52202/079017-1239. URL https://proceedings.neurips.cc/paper_files/paper/2024/file/4540d267eeec4e5dbd9dae9448f0b739-Paper-Conference.pdf.

Wang, J., Zhao, S., Luo, Z., Zhou, Y., Jiang, H., Li, S., Li, T., and Pan, G. CBramod: A criss-cross brain foundation model for EEG decoding. In *The Thirteenth International Conference on Learning Representations*, 2025a. URL https://openreview.net/forum?id=NPNUHgHF2w.

Wang, J., Zhao, S., Luo, Z., Zhou, Y., Li, S., and Pan, G. Eegmamba: An eeg foundation model with mamba. *Neural Networks*, 192:107816, 2025b. ISSN 0893-6080. doi: https://doi.org/10.1016/j.neunet.2025.107816. URL https://www.sciencedirect.com/science/article/pii/S0893608025006963.

You, Z., Guo, Y., Zhang, X., and Zhao, Y. Virtual electroencephalogram acquisition: A review on electroencephalogram generative methods. *Sensors*, 25(10), 2025. ISSN 1424-8220. doi: 10.3390/s25103178. URL https://www.mdpi.com/1424-8220/25/10/3178.

Zhao, T., Cui, Y., Ji, T., Luo, J., Li, W., Jiang, J., Gao, Z., Hu, W., Yan, Y., Jiang, Y., and Hong, B. Vaeeg: Variational auto-encoder for extracting eeg representation. *NeuroImage*, 304:120946, 2024. ISSN 1053-8119. doi: https://doi.org/10.1016/j.neuroimage.2024.120946. URL https://www.sciencedirect.com/science/article/pii/S1053811924004439.

Zhou, Y., Wu, J., Ren, Z., Yao, Z., Lu, W., Peng, K., Zheng, Q., Song, C., Ouyang, W., and Gou, C. CSBrain: A cross-scale spatiotemporal brain foundation model for EEG decoding. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2026. URL https://openreview.net/forum?id=agcXjEHmyW.

A. Foundation models used for comparisons

- a paragraph for each model describing the main contribution of the architecture / method applied.

B. Ablation study on inductive bias on spatial positionning of channels

C. Computational analysis

-

D. Implementation details